ALI AKARMA

AI Safety & Agentic AI Researcher

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RESEARCH INTERESTS

Trustworthy agentic AI systems, AI safety and alignment, constrained multi-agent reinforcement learning (C-Dec-POMDP, Lagrangian CTDE), AI governance and oversight architectures, adversarial robustness of LLM-based autonomous pipelines, blockchain-anchored audit mechanisms for safety-critical deployments, and digital twin frameworks for smart city infrastructure.

EDUCATION

B.S. Information Technology — Islamic University of Madinah, Saudi Arabia *2023 -- 2027*

- GPA: 4.29 / 5.00 — *Merit-based fully funded scholarship (competitive international selection)*
- Relevant coursework: Machine Learning, Algorithms & Data Structures, Probability & Statistics, Database Systems, Software Engineering
- Thesis interest: Safety-aligned multi-agent orchestration under Byzantine constraints

RESEARCH EXPERIENCE

Research Intern — King Fahd University of Petroleum and Minerals (KFUPM) *Jun 2026 -- Present*

- Investigating AI decision-making verification for wildfire suppression and mitigation strategies in Middle East environments using reinforcement learning and multi-agent systems.
- Developing evaluation protocols for safety, reliability, and trustworthiness of autonomous agent policies under distributional shift and adversarial conditions.

Undergraduate AI Researcher — Islamic University of Madinah *Jan 2025 -- Present*

- Designed constrained multi-agent RL architectures (C-Dec-POMDP, Lagrangian CTDE) for safety-critical disaster prediction, achieving 81.5 reward with 2.3% safety violation rate across six baselines on meteorological data from the Saudi National Center for Meteorology.
- Evaluated adversarial prompt-injection and jailbreak failure modes in autonomous LLM-agent pipelines; built a LangChain prompt-injection defence middleware with three detection modes and four policy strategies.
- Led development of governance-constrained agentic AI framework for wildfire monitoring with human-supervised governance framework for safety-critical AI systems (first-author, ICETAS 2026).
- Built 10+ open-source research prototypes with reproducible pipelines, pytest suites, and Zenodo-archived datasets, spanning healthcare (AgHealth+, ADAPT), disaster response (LagrangianCTDE), smart cities (AquaAgent), and finance (FinNutriAgent).

GRANT-FUNDED RESEARCH

King Salman Center for Disability Research (KSCDR) — *Research Group No. KSRG-2026-321*

- **ADAPT**: An agentic AI framework for disability-inclusive nutrition and healthcare monitoring. *Published in Journal of Disability Research*.

  DOI: 10.57197/JDR-2026-0830 | GitHub: github.com/aliakarma/ADAPT
- **FedAgent-Chain**: A secure federated and agentic AI framework for multilingual disability-inclusive employment in AI cities. Published in *MDPI Smart Cities*.

  DOI: 10.3390/smartcities9070106 | GitHub: github.com/aliakarma/fedagent-chain

SELECTED PUBLICATIONS

Google Scholar: 66 citations, h-index 5, i10-index 2.

First-Author Papers

- **Akarma, A.**, Syed, T.A., Jan, S., Muneer, H., Jilani, A.K. *"Governance-Constrained Agentic AI: Blockchain-Enforced Human Oversight for Safety-Critical Wildfire Monitoring,"* **ICETAS 2026**. arXiv:2604.04265
- **Akarma, A.**, Syed, T.A., Jilani, A.K., Jan, S., Muneer, H., Khan, M.A., Yu, C. *"Agents for Agents: An Interrogator-Based Secure Framework for Autonomous Internet of Underwater Things,"* **ICETAS 2026**. arXiv:2604.04262

- **Akarma, A.**, Syed, T.A., Hameed, D., Muneer, H. *"Improving Disability Support and Institutional Performance With AI,"* Book Chapter, **IGI Global**. 10.20944/preprints202608.1878.v1
- Syed TA, **Akarma A**, Alatify A, Naqash MT, Alqurashi A. "Agentic AI-enhanced digital twins for Smart City civil infrastructure: A secure, autonomous and auditable management framework" **PLOS ONE**. 10.1371/journal.pone.0353610 (Co-First Author)

Journal Articles

- Jan, S., **Akarma, A**., Syed, T.A. *et al.* "*EAGF: a four-pillar ethical AI governance framework for trustworthy cybersecurity in 5G renewable energy IoT systems*". ***Scientific Reports*** 2026. 10.1038/s41598-026-63383-5.
- Syed, T. A., **Akarma, A.**, Naqash, M. T., Hameed, D., Kamal, S., & Formisano, A. *"Agentic AI for Climate-Resilient Cities: A PRISMA-Guided Review and Digital Twin Framework,"* **MDPI Sustainability** (Accepted).
- Syed, T. A., Alshahrani, A., **Akarma, A.**, Khan, S., Nauman, M., Lee, I. E., ... & Ullah, A. *"FinNutriAgent (FNA): An Agentic AI for Budget and Nutrition Planning,"* **ETASR**, 2026. 10.48084/etasr.15640

Conference Papers

- Syed, T. A., Alshahrani, A., Ullah, A., **Akarma, A.**, Khan, S., Nauman, M., & Jan, S. *"FinAgent: Agentic AI for Personal Finance and Nutrition,"* **IEEE ICCA 2025**. 10.1109/ICCA66035.2025.11430760
- Jan, S., Razzaqi, H. A., **Akarma, A.**, & Belgaum, M. R *"Blockchain-Monitored Agentic AI Architecture for Trusted Perception-Reasoning-Action Pipelines,"* **IEEE ICCA 2025**. 10.1109/ICCA66035.2025.11430865

OPEN-SOURCE RESEARCH SOFTWARE

All repositories include reproducible experimental pipelines and Zenodo-archived datasets.

- **LagrangianCTDE** — Constrained MARL for disaster response (PyTorch, MAPPO, Lagrangian safety constraints). (Repository)
- **Prompt Injection Defense** — LangChain middleware with 3 detection modes and 4 policy strategies. (Repository)
- **AquaAgent** — MAPPO-based leak detection for water networks with EPANET 2.2 digital twin. (Repository)
- *Full list of 20+ repositories at* github.com/aliakarma*, each with Zenodo DOIs for citability.*

TECHNICAL SKILLS

Languages & Frameworks: Python, Java, SQL; PyTorch, NumPy, scikit-learn, Hugging Face Transformers, PuLP

AI & RL Methods: Multi-agent RL (MAPPO, CTDE), C-Dec-POMDP, Lagrangian constrained optimization, LLM orchestration, adversarial robustness, digital twin simulation

Safety & Governance: Failure-mode analysis, policy-aligned control, blockchain audit trails, prompt-injection defense, SHAP/LIME explainability

Research Tools: LaTeX, Git, Linux, Jupyter, EPANET 2.2, REST APIs, Zenodo, PRISMA systematic reviews

AWARDS & CERTIFICATIONS

- **Merit-Based Fully Funded Scholarship** — Islamic University of Madinah (competitive international selection, 2023)
- **Certificate of Appreciation (Research Excellence)** — ICETAS 2026, for research contribution on agentic AI governance
- **Certificate of Appreciation** — ICBDT 2025, for presented research contributions
- **Mindware: Critical Thinking for the Information Age** — University of Michigan (Coursera), 2024
- **Computational Thinking for Problem Solving** — University of Michigan (Coursera), 2024

LANGUAGES & SERVICE

Languages: Urdu (native), English (professional proficiency), Arabic (professional proficiency)

Service: Volunteer, Masjid Al-Nabawi, Madinah (Ramadan 2024, 2025) — visitor assistance during peak pilgrimage seasons.